

WealthWise Project Workflow Guide

How each role works through the banking and wealth-management platform.

Seed Users and Responsibilities

Role	Seed login	Main responsibility
ADMIN	admin@bank.com / admin123	Creates/manages users, opens/closes bank accounts, sees audit logs, can access all business modules where allowed.
ACCOUNTHOLDER	holder@bank.com / holder123	Uses accounts, beneficiaries, transfers, schedules, loans, fixed deposits, portfolios, EMIs, and notifications.
OPERATIONS	ops@bank.com / ops123	Views account/reporting operations and can reverse transfers.
LOANOFFICER	loan@bank.com / loan123	Reviews, approves/rejects loan applications, disburses loans, monitors EMIs and loan status.
RELATIONSHIPMANAGER	rm@bank.com / rm123	Manages portfolios, holdings, fixed-deposit review/maturity flows.
COMPLIANCE	compliance@bank.com / comp123	Handles KYC verification/rejection and AML investigation/clearance.

System Architecture Flow

Request flow: Controller -> Service Interface -> Service Implementation -> Repository -> Entity. DTOs carry API data across layers. Shared services such as LedgerService, NotifierService, and AuditTrailService connect modules safely.

1. Authentication and Security Flow

? Every user starts by calling POST /api/auth/login with email and password. The API returns token, role, and userId.

? All secured calls use Authorization: Bearer . The JWT filter validates signature, expiry, current user status, and role consistency.

? Locked or inactive users cannot continue using old tokens because the filter checks the current database user on every secured request.

? CORS is open for http://localhost:3000 and OPTIONS preflight is allowed.

2. Admin Onboarding and Account Creation Flow

? Admin logs in, then creates users through POST /api/users/create. This endpoint is ADMIN-only.

? To open a bank account, admin calls POST /api/bank-accounts with customerId, accountType, opening balance, and interest rate.

? The service validates that customerId belongs to an existing active ACCOUNTHOLDER. It rejects non-account-holder users and inactive/locked users.

? The client does not control accountNumber, openDate, or status. The backend generates accountNumber as account-type prefix + yyyyMMdd + six random digits: SB for Savings, CA for Current, SA for SalaryAccount.

? New accounts start as Active and can be closed only by ADMIN using PATCH /api/bank-accounts/{id}/close.

3. Account Holder Daily Banking Flow

- ? Account holder logs in and views accounts using /api/bank-accounts endpoints.
- ? Customer can add external or other-customer beneficiaries. They cannot add their own account as a beneficiary.
- ? Beneficiary creation validates required fields, active account-holder customer, and duplicate account numbers. Previously deleted beneficiaries can be reactivated.
- ? Fund transfer requires different source/destination accounts, positive amount, active source and destination accounts, and verified KYC for sender.
- ? Ledger updates debit and credit balances transactionally, creates daily account-statement rows, raises AML flags for large transfers, notifies both customers, and records audit trail.
- ? Scheduled payments must use the customer-own active source account and active beneficiary owned by the same customer. Creation always starts Active; cancelled schedules cannot be paused.

4. Loan and Credit Flow

- ? Account holder submits a loan application with customerId, loan type, amount, tenure, and purpose.
- ? Loan officer moves the application to under-review, then approves or rejects it.
- ? Only approved and not-yet-disbursed applications can be disbursed into the customer account.
- ? Loan account creates EMI schedules. EMI payment reduces outstanding balance. Overdue and summary endpoints support monitoring.
- ? Loan officer can mark a loan account NPA or close it.

5. Investment and Portfolio Flow

- ? Account holder or relationship manager creates fixed deposits. FD creation debits the linked active bank account through the ledger.
- ? FD can mature or close prematurely. Maturity credits the account through the ledger.
- ? Relationship manager creates portfolios and adds holdings. Holdings are nested under portfolios, not standalone endpoints.
- ? Holding redemption changes holding status and updates portfolio behavior through the portfolio service.

6. KYC, Compliance, and AML Flow

- ? Compliance creates KYC records for customers and then verifies or rejects records.
- ? Fund transfer checks sender KYC; transfers are blocked until sender KYC is Verified.
- ? AML flags can be created manually, investigated, and cleared. Large transfers automatically create AML flags.

7. Reporting and Notification Flow

- ? Operations/admin generate and view banking reports by scope and summary.
- ? Notifications are created by business events or directly through the notifications API. Users can list unread notifications, count unread messages, mark one read, or mark all read.

8. End-to-End Happy Path

- ? Admin logs in -> creates account-holder user -> opens bank account for that account holder.
- ? Compliance logs in -> submits and verifies KYC for the account holder.
- ? Account holder logs in -> adds beneficiary -> initiates transfer -> checks statement -> creates scheduled payment.
- ? Loan officer logs in -> reviews and approves loan -> disburses loan -> account holder pays EMI.
- ? Relationship manager logs in -> creates portfolio/adds holdings or manages FD maturity.
- ? Operations logs in -> reviews reports and may reverse transfers when required.

Important Business Rules

Area	Rule
Bank account creation	Only ADMIN can open/close bank accounts. Account number is generated by backend, not supplied by client. Customer must be active ACCOUNTHOLDER.
Account number format	Savings: SB + yyyyMMdd + six random digits. Current: CA + yyyyMMdd + six random digits. SalaryAccount: SA + yyyyMMdd + six random digits.
Transfers	Source and destination must be different active accounts; amount must be positive; sender KYC must be Verified; large transfers raise AML flags.
Beneficiaries	Required fields are validated; customer must be active account-holder; customer cannot add own account as beneficiary; deleted duplicate can reactivate.
Scheduled payments	Source account must belong to customer and be active; beneficiary must belong to customer and be active; creation always starts Active.
Ledger	Debit/credit reject zero or negative amounts and inactive accounts; ledger writes running daily statements.
JWT security	Every secured request checks token validity, current user status, and role consistency.